Trump Tweet Market Impact Classifier

Solution Overview & Results

1 Problem

Given a political tweet, the system must answer two questions:

- Is this tweet market-relevant?
- If so, in which direction does it move each of seven financial assets (gold, equities, BTC, crude oil, wheat, EUR/USD, 2Y Treasury) at the 1m, 5m, and 10m horizons?

2 Approach

A single tweet carries several independent signals: who is mentioned, what sentiment it conveys, what kind of event it describes, when it was posted, and how that event propagates through connected markets. Rather than fine-tuning one large model on raw text, the pipeline decomposes these signals into five structured feature layers, then feeds the combined ~130-feature vector into a LightGBM ensemble.

3 Five-Layer Feature Pipeline

Layer 1 - Named Entity Recognition (GLiNER)

Extracts persons, countries, commodities, organizations, and policy terms using zero-shot NER. Each entity is mapped to one or more target asset classes via a curated lookup table. Falls back to keyword matching when GPU is unavailable. Produces 14 features.

Layer 2 - Financial Sentiment (FinBERT)

FinBERT is pre-trained on financial corpora and outputs positive/negative/neutral probabilities plus a compound score in [-1, 1]. More domain-calibrated than general-purpose sentiment models. Produces 8 features.

Layer 3 - Event Detection (DistilBERT + Trainable Head)

A frozen DistilBERT encoder feeds a lightweight two-layer MLP head trained on pseudo-labeled tweets. Classifies 13 event types (trade war, sanctions, monetary policy, geopolitical conflict, election, etc.). A hybrid overlay of 100+ regex rule patterns boosts recall on rare event types. Produces 27 features.

Layer 4 - Context Enrichment

Extracts temporal signals (trading session, hour-of-day, day-of-week), tweet stylistic signals (caps ratio, urgency, URL presence), and thread position features. Optionally fetches live macro news via Tavily Search API and extracts thematic flags. Produces 18 features.

Layer 5 - Entity-Event-Asset Graph Reasoning

Builds a directed weighted graph per tweet using 97 domain-knowledge transmission rules. Signal propagates from entity nodes through event nodes to asset nodes. Sentiment compound score modulates each asset's net signal by +/-30%. Produces 21 features.

The ~130 assembled features are passed to a LightGBM ensemble consisting of:

- 1 binary relevance classifier - market-relevant vs. not relevant
- 21 multiclass direction classifiers - 7 assets x 3 timeframes, each predicting -1 / 0 / +1

4 Key Design Decisions

Modular layers over end-to-end fine-tuning

With ~2,700 labeled tweets, fine-tuning a large transformer risks overfitting. Decomposing into interpretable layers lets each component specialise while keeping the final learner small and fast.

Graph transmission rules

Structured financial domain knowledge is encoded directly as rules (e.g. US-China trade war -> equities bearish, gold bullish). This strong prior compensates for the limited training data.

Hybrid event detection

The trained DistilBERT head handles common event types; 100+ regex patterns cover rare categories (military, crypto policy) that appear too infrequently to learn from data alone.

Temporal train/test split

The oldest 80% of tweets train the models; the most recent 20% are held out. This prevents leakage and mirrors real deployment conditions.

LightGBM for final prediction

Fast training, native class-weight balancing, and interpretable feature importances (used to generate per-prediction reasoning strings). Gradient-boosted trees outperform deep models on this tabular feature set at this data scale.

5 Results

Evaluated on 667 held-out test tweets via strict temporal split (no test data seen during training).

Classifier	Metric	Score
Relevance Classifier	Accuracy	**85.31%**
Relevance Classifier	ROC-AUC	**89.72%**
Relevance Classifier	Recall	**75.35%**
Relevance Classifier	CV Accuracy (5-fold)	**95.47%**
Relevance Classifier	CV F1 (5-fold)	**95.54%**

The relevance classifier is the primary production-grade component, correctly identifying market-moving tweets with 85% accuracy and 89.7% ROC-AUC. The directional models are directionally informative but reflect the inherent difficulty of predicting short-horizon price moves from tweet text alone.